

Supplementary Information for “Regime maps for exact quantum compilation separate expressivity, certificates and feature transfer”

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1 Formal model definitions

1.1 Pauli notation

An n -qubit Pauli word modulo global phase is represented by $(x, z) \in \mathbb{F}_2^{2n}$. Multiplication is vector addition. Its support is $\text{wt}(x, z) = |\{q : (x_q, z_q) \neq (0, 0)\}|$, and its commutation parity with (x', z') is $\langle (x, z), (x', z') \rangle = x \cdot z' + z \cdot x' \pmod{2}$. All theorem statements below quantify over every n and every target tuple for which the stated feasible set is nonempty.

1.2 Shared-tag paired-block model

An instance is six nonidentity targets P_{jk} , with $j \in \{0, 1, 2\}$ and $k \in \{0, 1\}$. A configuration consists of six nonidentity frames R_{jk} , a tag S , and a central index $c_j \in \{0, 1\}$ for each block, subject to

$$\langle R_{j0}, R_{j1} \rangle = 1, \quad (1)$$

$$\langle S, R_{j0} \rangle = \ell_0, \quad \langle S, R_{j1} \rangle = \ell_1, \quad (2)$$

$$(\ell_0, \ell_1) \in \{(0, 1), (1, 0)\}, \quad (3)$$

with the same (ℓ_0, ℓ_1) for all three blocks. Write $T_{jk} = P_{jk}R_{jk}$. For local Pauli letters define

$$F_3(a, b, c) = \begin{cases} 1, & a = b = c \neq I, \\ \mathbf{1}[a \neq I] + \mathbf{1}[b \neq I] + \mathbf{1}[c \neq I], & \text{otherwise.} \end{cases} \quad (4)$$

The structural objective is

$$\begin{aligned} C_{\text{shared}} = & \sum_{j=0}^2 \left[2(\text{wt}(R_{j,c_j}) - 1) + 4(\text{wt}(R_{j,1-c_j}) - 1) \right] + 2 \text{wt}(S) \\ & + \sum_{q=1}^n \sum_{k=0}^1 F_3(T_{0k}[q], T_{1k}[q], T_{2k}[q]). \end{aligned} \quad (5)$$

The subtracted one in each frame term is a constant normalization over feasible configurations. The support coordinate is $\max_{j,k} \text{wt}(R_{jk})$.

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1.3 Rank-two dependent-triple model

An instance contains two ordered target triples P_{bk} , $b \in \{A, B\}$ and $k \in \{0, 1, 2\}$. A configuration selects a relative permutation π_B , a central branch $c_b \in \{0, 1, 2\}$ in each block, independent frames R_{b0}, R_{b1} satisfying $\langle R_{b0}, R_{b1} \rangle = 1$, dependent frames $R_{b2} = R_{b0}R_{b1}$, and two shared tags S_0, S_1 . Define

$$\lambda(R) = 2\langle S_0, R \rangle + \langle S_1, R \rangle \in \{0, 1, 2, 3\}. \quad (6)$$

Feasibility requires

$$(\lambda(R_{A0}), \lambda(R_{A1})) = (\lambda(R_{B0}), \lambda(R_{B1})) = (a, b), \quad a, b \in \{1, 2, 3\}, \quad a \neq b. \quad (7)$$

The dependent label is then $a \text{ xor } b$, the third nonzero label. With π_A the identity, the objective is

$$C_{\text{rank2}} = \sum_{b \in \{A, B\}} \left[t_c (\text{wt}(R_{b, c_b}) - 1) + t_{nc} \sum_{k \neq c_b} (\text{wt}(R_{bk}) - 1) + t_r \sum_{k=0}^2 \text{wt}(P_{b, \pi_b(k)} R_{bk}) \right] + t_{\text{tag}} (\text{wt}(S_0) + \text{wt}(S_1)). \quad (8)$$

The unit structural objective is $(t_c, t_{nc}, t_{\text{tag}}, t_r) = (2, 4, 2, 1)$. The support coordinate is $\max_{b, k \in \{0, 1\}} \text{wt}(R_{bk})$.

1.4 Six-term linear-combination model

An instance is a formal batch $(p_1, \dots, p_6)$ of nonidentity n -qubit Pauli words; repeated words are allowed. A configuration is a set partition G of $\{1, \dots, 6\}$, a factoring bit φ_g for every block, and either shared or dedicated index ancillas. For a block g of size m , let $b(m) = \lceil \log_2 m \rceil$, with $b(1) = 0$. Its common factor F_g contains the local letter shared nontrivially by every p_i , $i \in g$, and identity elsewhere; $r_i = p_i F_g$ modulo phase. Let $f = 1$ when G has at least two blocks and $f = 0$ otherwise. The selection contribution of g is

$$\text{SEL}_g = \begin{cases} \sum_{i \in g} (f + b(m) + 1) \text{wt}(p_i), & \varphi_g = 0, \\ (f + 1) \text{wt}(F_g) + \sum_{i \in g} (f + b(m) + 1) \text{wt}(r_i), & \varphi_g = 1. \end{cases} \quad (9)$$

Writing $d(1), \dots, d(6) = (0, 0, 1, 2, 4, 6)$ for the frozen balanced-tree depth sums,

$$\text{PREP}(G) = \mathbf{1}[|G| \geq 2] (2|G| - 3) + \sum_{g: |g| \geq 2} [(|g| - 1)(1 + f) + d(|g|)], \quad (10)$$

$$\text{WIDTH}(G) = \mathbf{1}[|G| \geq 2] |G| + \begin{cases} \max_g b(|g|), & \text{shared,} \\ \sum_g b(|g|), & \text{dedicated.} \end{cases} \quad (11)$$

The exact family cost is the minimum of $\sum_g \text{SEL}_g + \text{PREP} + \text{WIDTH}$ over all 203 partitions, factoring bits and ancilla choices. The reference is the six-singleton partition, with cost $C_U = 2 \sum_i \text{wt}(p_i) + 15$.

2 All-size normalization proofs

2.1 Shared-tag support-two proof

Fix a feasible configuration and a frame $R = R_{jk}$ with support $w \geq 3$; let R' be its anticommuting partner. At each $q \in \text{supp}(R)$ assign

$$(\alpha_q, \beta_q) = (\langle R[q], R'[q] \rangle, \langle S[q], R[q] \rangle) \in \mathbb{F}_2^2. \quad (12)$$

The total α parity is one. Any such multiset of size at least three contains either a $(0,0)$ singleton or an equal pair. Indeed, if there is no zero and no repetition, it is a subset of the three nonzero vectors; the only size-three case has zero total α , a contradiction. Removing the indicated proper subset Q , with $1 \leq |Q| \leq 2$, therefore changes neither total α nor total β . Zeroing R on Q preserves block anticommutation and the shared tag syndrome, leaves the other two blocks unchanged, and needs no tag repair.

It remains to compare cost. For each removed local letter, central or noncentral frame support refunds respectively two or four units. Exhaustive evaluation of the local factor F_3 in Eq. (5) covers the removed nonidentity letter, its partner and tag letters, the raw target letter, the two other block letters, the branch position and both multipliers. Across all 18,432 cases the increase in F_3 is at most the frame refund. Thus the edit is non-increasing in cost and strictly decreases total frame support. Repeating it terminates with support at most two. The four two-letter class patterns that do not contain a removable zero-sum proper subset explain why this argument stops at two rather than one.

2.2 Rank-two support-one proof

The support-one proof begins from a support-two normal form for the same feasible family. The precursor certificate uses a syndrome-preserving relabel-and-delete grammar. Its full local action space contains 612 support-three profile cases; 21 are unsafe before imposing Eq. (7), and none of those 21 is feasible after the complete label predicate is applied. Hence every optimum has a support-two representative. The packaged checker reconstructs these profiles from local Pauli descriptors and verifies the acceptance filter; the 144-case support-two control retains 36 feasible obstructions to that older edit grammar, so the precursor alone does not claim support one.

Now take any support-two representative. Each anticommuting pair has exactly one qubit at which its two local letters anticommute; call it the core. Delete every non-core active column, align each remaining local basis at its core, and reconstruct the tags only after both blocks have been localized. The complete local bounds are:

1. deleting a commuting active column refunds at least four objective units (and deleting an anticommuting column refunds at least seven);
2. changing an anticommuting local basis at a fixed core changes the restoration term by at most three while leaving the frame contribution equal to ten;
3. on a shared core, feasibility forces the two blocks to use the same local basis, and the canonical dual tag costs four;
4. on distinct cores, the canonical dual tags cost eight; if both blocks were already support one the original tag costs at least eight, while otherwise one non-core deletion from each affected side supplies the additional cost.

The numerical inequalities $4 > 3$, $4 + 4 \geq 8$ and the tag lower bounds close the same-core and distinct-core cases. The edit preserves Eq. (7) by construction and never increases Eq. (8). Both independent generators end on one core. Support zero is infeasible because $\langle I, I \rangle = 0$, so the anticommutation constraint cannot be met. Hence the intrinsic support number is one.

Table 1: Finite obligations supporting the two all-size normalization results. These enumerations discharge local cases inside the analytic compositions; they are not samples from a deployment population.

Model	Obligation	Rows	Outcome
Shared-tag Pauli	local exchange inequality	18,432	0 violations
Shared-tag Pauli	parity-class tuples	43,688	exact support-two boundary; 0 failures for support 3–8
Rank-two dependent triple	precursor action profiles	612	21 unsafe before full acceptance; 0 accepted unsafe
Rank-two dependent triple	deletion cases	2,880	worst changes -4 and -7
Rank-two dependent triple	core alignment	6,912	maximum increase $+3$
Rank-two dependent triple	same-core tag rigidity	576	0 different-basis counterexamples
Rank-two dependent triple	distinct-core tag lower bound	9,216	minimum tag cost 8

3 Objective-indexed certificate

For a deleted commuting column, let (r_c, r_{nc}, Δ_r) record refunded central support, refunded noncentral support and restoration increase. The complete 1,728-row resource domain has two undominated worst vectors, $(0, 2, 2)$ and $(1, 1, 2)$. Core alignment may add a further $3t_r$, while moving the two tag words to a second core may add $2t_{\text{tag}}$. Requiring each worst refund to cover either obligation gives

$$2t_{nc} \geq 5t_r, \quad t_c + t_{nc} \geq 5t_r, \quad (13)$$

$$2t_{nc} \geq 2t_r + 2t_{\text{tag}}, \quad t_c + t_{nc} \geq 2t_r + 2t_{\text{tag}}. \quad (14)$$

For $(2, 4, 2, 1)$ the left-minus-right margins in displayed order are $(3, 1, 2, 0)$. For $(3, 5, 2, 1)$ they are $(5, 3, 4, 2)$. Thus the first point lies on the certificate boundary and the second lies strictly inside. These inequalities are sufficient for this proof. Their violation provides no converse statement.

4 Six-term pair-gain equivalence

For $B \subseteq \{1, \dots, 6\}$, write $s(B) = \sum_{i \in B} \text{wt}(p_i)$ and let $f(B)$ count all-equal nonidentity columns. The gain of a factored block of size m has coefficients (A_m, B_m) multiplying $(f(B), s(B))$:

$$(A_m, B_m) = (0, 0), (4, -1), (10, -2), (14, -2), (23, -3) \quad (m = 1, \dots, 5). \quad (15)$$

For $m = 6$ the gain is $23f(B) - 2s(B) + 1$. A partition contributes the sum of its block gains plus the shape constant shown in Table 2.

Let $g(e) = 4f(e) - s(e)$ for each pair. Condition P_0 consists of $g(e) \leq 0$ for every pair, $g(e) + g(f) + 1 \leq 0$ for every two disjoint pairs, and $g(e) + g(f) + g(h) + 2 \leq 0$ for every perfect matching. Summing pair clauses within a block yields $s(B) \geq 6f(B)$ for

triples, $s(B) \geq 8f(B)$ for quads, $s(B) \geq 10f(B)$ for quints, and $s(B) \geq 12f(B) + 2$ for the six-term block. Substitution makes every non-pair shape gain nonpositive. The three multi-pair shapes are exactly the three clauses of P_0 . Conversely, violation of any clause makes its corresponding pair partition strictly cheaper than the singleton reference. This proves $C_F = C_U$ if and only if P_0 for every admitted batch and every n .

Table 2: Complete shape ledger for the six-term equivalence. T_m is the affine gain of one factored block of size m ; g_i denotes a pair gain.

Shape	Partitions	Exact gain form	Bound under P_0
$1 + 1 + 1 + 1 + 1 + 1$	1	0	equality
$2 + 1 + 1 + 1 + 1$	15	g_1	pair clause
$2 + 2 + 1 + 1$	45	$g_1 + g_2 + 1$	two-disjoint-pair clause
$2 + 2 + 2$	15	$g_1 + g_2 + g_3 + 2$	perfect-matching clause
$3 + 1 + 1 + 1$	20	$T_3 - 1$	$T_3 \leq 0$
$3 + 2 + 1$	60	$T_3 + g_1$	both nonpositive
$3 + 3$	10	$T_3 + T'_3$	both nonpositive
$4 + 1 + 1$	15	$T_4 - 1$	$T_4 \leq 0$
$4 + 2$	15	$T_4 + g_1$	both nonpositive
$5 + 1$	6	$T_5 - 3$	$T_5 \leq 0$
6	1	$23f - 2s + 1$	at most $-f - 3$

5 State-preparation model and exact feature construction

A state is the full signed stabilizer group in canonical sorted binary form. The allowed gates are H , S , $S^\dagger$ and controlled-NOT with costs 1, 1, 1, 3. The exact referee is Dijkstra search from $|0 \cdots 0\rangle$ over the complete stabilizer-state graph. The deterministic reference synthesis processes qubits in ascending order, selects the lexicographically smallest admissible X pivot, uses the X elimination route when available, otherwise uses the forced signed- Z route, and applies the inverse elimination sequence for preparation. Sign repairs and controlled-NOT routes are therefore deterministic functions of the target state.

The 127 coordinates are ordered in three blocks. The 33 schedule/tensor coordinates comprise 13 base quantities and 20 trace quantities. The base quantities are five reference event counts, reference cost, binary X rank, tensor-factor count, the lower bound $r_X + 3(n - c)$, and four declared differences from these quantities. The trace block records maximum, first and last step costs, the maximizing index, cost descents and moment, high/zero-cost step counts, event maxima and sign/ Y co-occurrences, signed-pivot and pivot-weight summaries, Z -route and rank-drop counts, largest tensor-factor size and reference cost, the enlarged-schedule cost, and its improvement over the reference.

The 53 path coordinates comprise four gate totals; minimum, maximum, sum of squares and zero count for each of five per-qubit gate channels; eight directed controlled-NOT graph summaries; all 16 ordered adjacent gate-kind transition counts; and first kind, last kind, run count, maximum run and number of distinct kinds. The 41 state coordinates comprise the negative-sign count; positive/negative Pauli-weight histograms for weights zero through four; positive/negative Y -count histograms; positive/negative binary parity-class counts; and minimum, maximum, sum of squares and zero count of the per-qubit X , Y and Z column marginals. The archive gives the exact coordinate names, order and formulas and supplies every panel vector.

The complete $n \leq 3$ domain has 1,146 states and 1,109 feature cells, including 1,072 singletons. No cell is mixed, so a lookup recovers every in-domain label. The ratio

$1,109/1,146 = 0.967714$ shows that the representation gives little compression.

Table 3: Complete and prospective state-preparation counts retained in the analysis.

Domain	States	Reference compiler exact
$n = 1$ complete	6	5
$n = 2$ complete	60	28
$n = 3$ complete	1,080	189
$n = 4$ frozen panel	120	32

6 Frozen panel, transfer rule and shuffle null

Using NumPy’s default generator with seed 20260821, each candidate panel state is produced from $|0000\rangle$ by 24 draws. A draw chooses uniformly among $H, S, S^\dagger$, and controlled-NOT; a one-qubit gate receives a uniform qubit, while a controlled-NOT receives a uniform control and then a uniform distinct target. Duplicate final states are discarded until 120 unique states have been retained in generation order. The review materials provide the ordered list of canonical states. Exact four-qubit costs are then read from the 36,720-state Dijkstra graph.

The representation and in-domain success criterion were frozen before four-qubit evaluation. Panel index parity defines two folds. For each test state, the other fold supplies counts for an exactly equal 127-coordinate vector. A seen cell predicts the strict majority label, with a tie mapped to negative; an unseen cell also predicts negative. Coverage is the number of test states whose vector occurs in the other fold. The 120 panel states form 119 cells: 118 singletons and one pure doubleton, so the partition invariant is $118 \times 1 + 1 \times 2 = 120$. Only the two states in the doubleton are covered across parity folds. Both are classified correctly; all 32 errors occur among the 118 uncovered states and equal the panel’s 32 positive labels.

For the null, the feature vectors, state order, folds, lookup rule and negative default remain fixed. Python’s pseudorandom generator with seed 20260828 produces 200 independent permutations of the 120 observed labels. Each permuted label vector is evaluated by the identical two-fold procedure. If $E_0 = 32$ is the observed error count and E_b is permutation b ’s error count, the reported empirical probability is

$$p_{\leq} = \frac{1}{200} \sum_{b=1}^{200} \mathbf{1}[E_b \leq E_0]. \quad (16)$$

The null counts range from 30 to 33, have mean 32.41, and give $p_{\leq} = 0.51$. This descriptive randomization result says that the declared lookup shows no improvement under this null on this panel. It is not an equivalence test.

The earlier prospective forecast is a separate frozen analysis. Its 120 panel states, predicted labels and predicted costs were fixed before the four-qubit referee was invoked. It matched 100 regime labels and 67 exact costs. Later feature work does not revise those predictions.

7 Nearest-work subtraction and consequence

Table 4: Claim-by-claim nearest-work subtraction. The comparison identifies the residual mathematical object and its consequence; it is not a priority certificate.

Advance	Closest work	Difference in object, assumptions and guarantee	Consequence retained here
Typed exact-regime record	Instance-space analysis and algorithm selection [1, 2]	Prior work maps features to performance or choices; the typed record separately represents theorem, certificate, representation and transfer fields with explicit missing-value semantics.	Numerical fields cannot borrow authority across type, objective or domain.
Shared-tag support two	Pauli compilation and cluster diagonalization [3, 4]	Prior work supplies Pauli compilation constructions; the present object is the declared three-block shared-tag grammar and the guarantee is an all-size support-two normal form under Eq. (S4).	Search may be restricted to support two in this grammar; no hardware claim follows.
Rank-two intrinsic support one	Verified and exact circuit optimization [5, 6]	Prior work establishes correct rewrites or exact matching; the present guarantee is tight support one for the dependent-triple grammar, with an objective-indexed proof region.	A conserved proof ceiling of five is not intrinsic support.
Six-term pair-gain boundary	Unitary partitioning and Pauli optimization [7, 8]	Prior work motivates grouping and Pauli optimization; the present object fixes six terms, 203 encodings and one structural objective, and proves an if-and-only-if three-clause boundary.	Reference exactness is decidable from pair-derived gains in this family only.
Adverse feature transfer	Quantum compiler prediction and bounded optimal Clifford synthesis [9–11]	Prior work predicts compilation choices or computes bounded optima; the present study couples a complete finite feature partition to a frozen next-size exact referee and retains failure.	Finite determination does not imply next-size coverage or transfer.

The comparison supports a narrow consequence. Instance-space analysis and compiler prediction remain the closest frameworks for feature-conditioned performance, while verified and exact synthesis remain the closest frameworks for correctness and bounded optimality. The present exact results add model-specific normal forms and a pair-gain equivalence; the typed record prevents their theorem authority from being transferred to the finite feature study. It does not displace the cited frameworks or establish a universal compiler theory.

8 Retained adverse and conditional results

The first prospective size-transfer forecast was fixed before the exact four-qubit distances were opened. It matched 100 of 120 reference-exactness labels and 67 of 120 exact costs. Both axes failed their exactness criteria. These outcomes remain visible because later representation work does not retroactively change a prospective forecast.

A separate search tested whether support two became strictly beneficial outside the proved support-one region. After correcting the treatment of infeasible local cells, the complete frozen search still found zero strict witnesses among 211,248 candidates under four objectives. The result excludes those candidates only. It does not prove that the objective region is globally sharp.

The measured gap between syndrome rank and intrinsic support equals a local exchange margin on two models. The equality has no residual degrees of freedom with two points, and it fails for one model when the syndrome rank is recomputed under a different aligned rewrite. It is retained as a diagnostic observation, not a theorem.

Table 5: Adverse and conditional results that constrain interpretation.

Question	Exact result	Allowed interpretation
Prospective state-preparation forecast	100/120 regime labels and 67/120 exact costs matched	the frozen forecast failed; no success-rate generalization follows
Support increase outside the certificate region	211,248 candidates, 0 strict witnesses under four objectives	a frozen-domain negative only; the global boundary remains open
Rank versus intrinsic support	slack equals the measured exchange margin on two models	a two-point, rewrite-dependent observation rather than a law
Rich state representation	1,109 cells for 1,146 states; 1,072 singleton cells	finite-domain vocabulary existence, not compact structure or transfer

9 Reproduction order

The review materials support four layers of checking. First, inspect the exact result records. Second, run the separate verifier, which checks theorem fields, objective vectors, panel-partition arithmetic, table values and required claim wording without invoking a scientific calculation. Third, regenerate all tables and the per-state feature and prediction record. Fourth, where the declared Python dependencies are available, rerun the selected

scientific analyses and compare their deterministic outputs. These operations detect inconsistent values and implementation drift. They are same-team checks, not external replication or independent proof review.

References

- [1] John R. Rice. The algorithm selection problem. In *Advances in Computers*, volume 15, pages 65–118. Elsevier, 1976. DOI: [10.1016/S0065-2458\(08\)60520-3](https://doi.org/10.1016/S0065-2458(08)60520-3).
- [2] Kate Smith-Miles and Mario Andrés Muñoz. Instance space analysis for algorithm testing: Methodology and software tools. *ACM Computing Surveys*, 55(12):1–31, 2023. DOI: [10.1145/3572895](https://doi.org/10.1145/3572895).
- [3] Filipa C. R. Peres and Ernesto F. Galvão. Quantum circuit compilation and hybrid computation using pauli-based computation. *Quantum*, 7:1126, 2023. DOI: [10.22331/q-2023-10-03-1126](https://doi.org/10.22331/q-2023-10-03-1126).
- [4] Ewout van den Berg and Kristan Temme. Circuit optimization of hamiltonian simulation by simultaneous diagonalization of pauli clusters. *Quantum*, 4:322, 2020. DOI: [10.22331/q-2020-09-12-322](https://doi.org/10.22331/q-2020-09-12-322).
- [5] Kesha Hietala, Robert Rand, Shih-Han Hung, Xiaodi Wu, and Michael Hicks. A verified optimizer for quantum circuits. *Proceedings of the ACM on Programming Languages*, 5(POPL):1–29, 2021. DOI: [10.1145/3434318](https://doi.org/10.1145/3434318).
- [6] Raban Iten, Romain Moyard, Tony Metger, David Sutter, and Stefan Woerner. Exact and practical pattern matching for quantum circuit optimization. *ACM Transactions on Quantum Computing*, 3(1):1–41, 2022. DOI: [10.1145/3498325](https://doi.org/10.1145/3498325).
- [7] Artur F. Izmaylov, Tzu-Ching Yen, Robert A. Lang, and Vladyslav Verteletskyi. Unitary partitioning approach to the measurement problem in the variational quantum eigensolver method. *Journal of Chemical Theory and Computation*, 16(1):190–195, 2020. DOI: [10.1021/acs.jctc.9b00791](https://doi.org/10.1021/acs.jctc.9b00791).
- [8] Jennifer Paykin, Albert T. Schmitz, Mohannad Ibrahim, Xin-Chuan Wu, and A. Y. Matsuura. PCOAST: A pauli-based quantum circuit optimization framework. In *2023 IEEE International Conference on Quantum Computing and Engineering*, pages 715–726, 2023. DOI: [10.1109/QCE57702.2023.00087](https://doi.org/10.1109/QCE57702.2023.00087).
- [9] Nils Quetschlich, Lukas Burgholzer, and Robert Wille. Predicting good quantum circuit compilation options. In *2023 IEEE International Conference on Quantum Software*, pages 43–53, 2023. DOI: [10.1109/QSW59989.2023.00015](https://doi.org/10.1109/QSW59989.2023.00015).
- [10] Nils Quetschlich, Lukas Burgholzer, and Robert Wille. MQT Predictor: Automatic device selection with device-specific circuit compilation for quantum computing. *ACM Transactions on Quantum Computing*, 6(1):1–26, 2025. DOI: [10.1145/3673241](https://doi.org/10.1145/3673241).
- [11] Sergey Bravyi, Joseph A. Latone, and Dmitri Maslov. 6-qubit optimal Clifford circuits. *npj Quantum Information*, 8:79, 2022. DOI: [10.1038/s41534-022-00583-7](https://doi.org/10.1038/s41534-022-00583-7).